

Abstract

AI & IoT Based Malpractice and Cheating Detection System for Online Exams is an intelligent examination monitoring system designed to improve the security, reliability, and fairness of online examinations. With the rapid growth of online education and remote assessments, traditional online examination platforms face several challenges, including unauthorized communication, tab switching, use of additional devices, suspicious movements, and other forms of malpractice. The proposed system combines Artificial Intelligence (AI), Internet of Things (IoT), computer vision, sensors, and real-time web-based monitoring to detect and respond to suspicious activities during examinations. The system provides separate Admin and Student modules. Students can securely log in and attend online examinations through a web-based examination interface, while administrators can monitor examination sessions and receive alerts when suspicious activities are detected. The software component monitors activities such as tab switching, window minimization, multiple-screen usage, and abnormal examination behavior. When a violation is identified, the system can generate an alert and, depending on the configured rules, automatically submit or terminate the examination. The IoT component uses devices such as an ESP32 microcontroller, PIR sensor, sound sensor, and ESP32-CAM to monitor the examination environment. The PIR sensor can identify movement, while the sound sensor can detect unusual sound activity. The ESP32-CAM provides visual monitoring of the examination area and can support AI-based analysis for identifying suspicious activities. The collected sensor information is transmitted to the web application through wireless communication, allowing administrators to observe the examination environment in real time. An AI-based detection layer can analyze visual and sensor data to identify potential malpractice patterns and classify activities according to their risk level. Instead of depending on a single sensor or detection method, the system combines multiple sources of information to improve monitoring reliability and reduce false alerts. A centralized dashboard displays the detected events and enables administrators to take appropriate action. The proposed system also maintains examination-related information and detection events using a database, providing a record of suspicious activities for later review. The combination of AI, IoT, web technologies, and real-time monitoring makes the system suitable for colleges, universities, training institutions, and other organizations conducting remote examinations.

Keywords :

Artificial Intelligence (AI), Internet of Things (IoT), Online Examination, Malpractice Detection, Cheating Detection, ESP32, ESP32-CAM, Computer Vision, Real-Time Monitoring, PIR Sensor

